```
    promise(promise const&) = delete;
    promise& operator=(promise const&) = delete;

    void swap(promise& ) noexcept;

    std::future<ResultType> get_future();

    void set_value(see description);
    void set_exception(std::exception_ptr p);
};
```

Std::Promise Default Constructor

Constructs a `std::promise` object.

Declaration

```
promise();
```

Effects

Constructs a `std::promise` instance with an associated asynchronous result of type `ResultType` that's not *ready*.

Throws

An exception of type `std::bad_alloc` if the constructor is unable to allocate memory for the asynchronous result.

Std::Promise Allocator Constructor

Constructs a `std::promise` object, using the supplied allocator to allocate memory for the associated asynchronous result.

Declaration

```
template<typename Allocator>
promise(std::allocator_arg_t, Allocator const& alloc);
```

Effects